

Unit-1

Introduction to DC circuits, series circuits, parallel circuits, series-parallel circuits, star/delta transformation.

Introduction to Electric Circuits

Electric Circuit: The path taken by the electric current to flow through an electrical load is called an electrical circuit.

A closed circuit in which dc current flows is called a dc circuit.

A closed circuit in which ac current flows is called a ac circuit.

Circuit Elements: They are of two types

(i) **Active Elements:** Active circuit elements are components that can generate, amplify, or control the flow of energy within a circuit.

Active elements are voltage and current sources which are able to supply energy to the circuit. Eg: Battery, signal generators.

Active elements are amplifiers which are able to amplify signals, increasing the strength of electrical signals. Eg: transistors, operational amplifiers.

Active elements are those which are able to control the flow of current and voltage in a circuit. Eg: diode, SCR

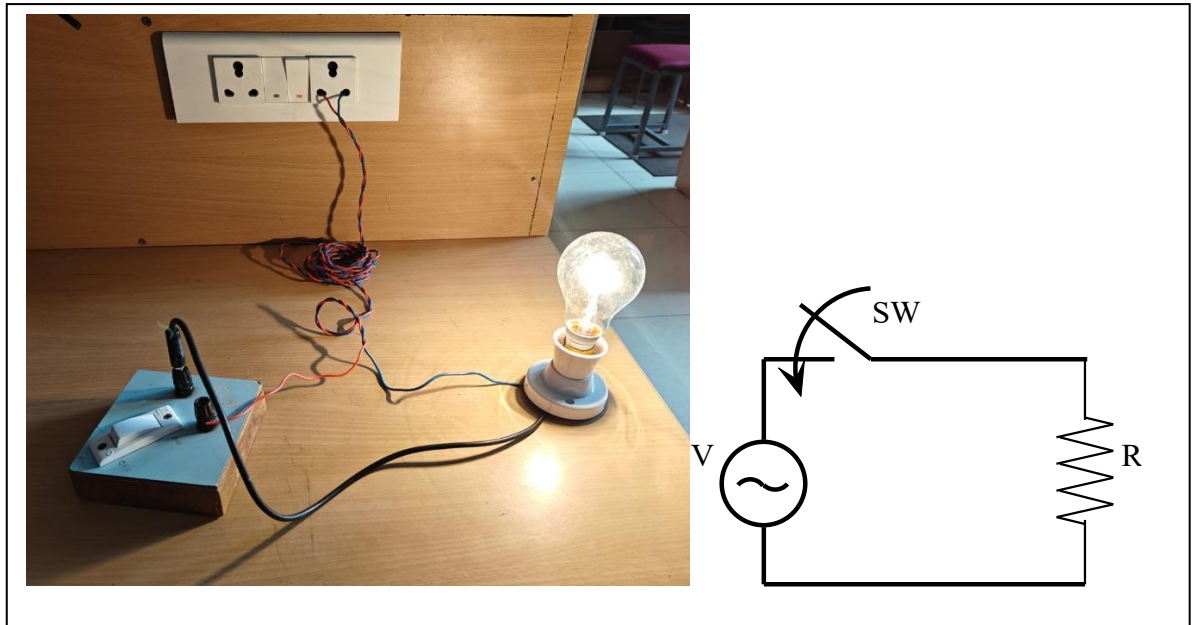
(ii) **Passive Elements:** Passive elements are those elements which either absorb or store or dissipate energy from the sources. Eg: Resistors, capacitors and inductors.

Examples can be put in the form of table

	Linear	Nonlinear
Passive	Resistor, Inductor, Capacitor, Mutual Inductor	Diode
Active	Voltage Source, Current Source, CCVS, CCCS VCCS, VCVS	LED, MOSFET, SCR, BJT

Note: Linear Elements exhibit linear i-v characteristics

Consider the electric circuit shown in the figure shown in Fig 1.1



Difference between electric circuit and electric network:

An electric network is any possible interconnection of electric circuit elements or branches. An electric circuit is a closed energized network. But a network need not be a closed energized circuit. The network shown in Fig 1.2 is an electrical network.

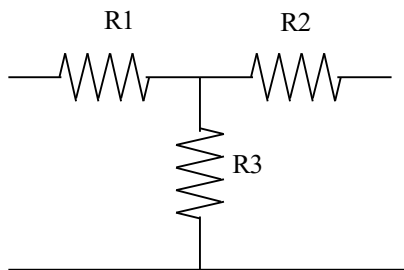


Fig 1.2 An example of an electrical network

Energy Sources: There are two types of sources of electrical energy. They are Voltage Source and Current Source as shown in Fig 1.3.

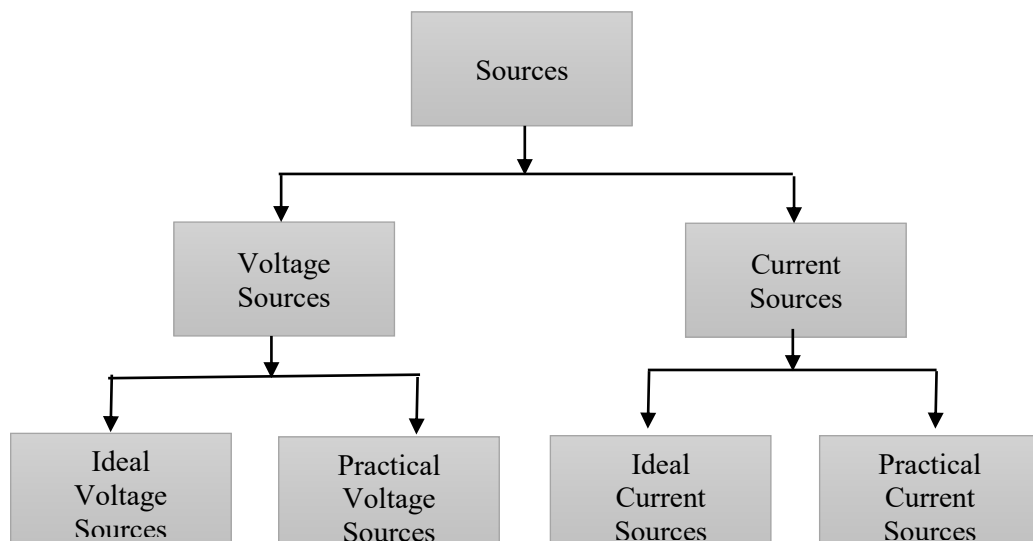


Fig 1.3 Electrical Energy Sources

1a. Ideal Voltage Sources: Ideal voltage source is defined as the energy source which gives constant voltage across the load terminals irrespective of the current drawn through its terminals and is shown in Fig 1.4

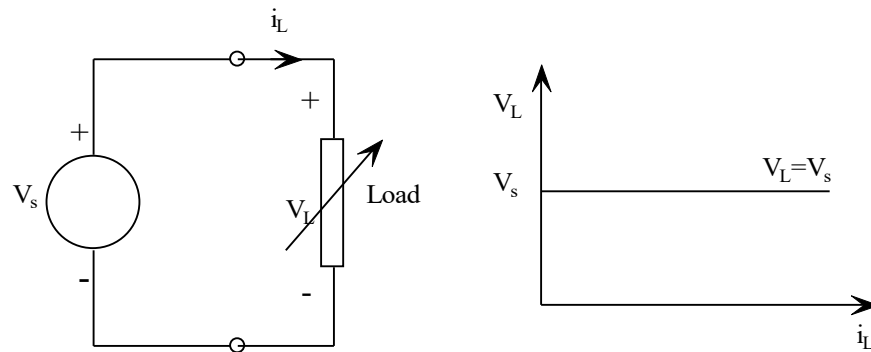


Fig 1.4 Ideal Voltage Source and its characteristic

1b. Practical Voltage source: In a practical voltage source, the voltage across the terminals of the source keeps falling as the current flowing through it increases as shown in Fig 1.5

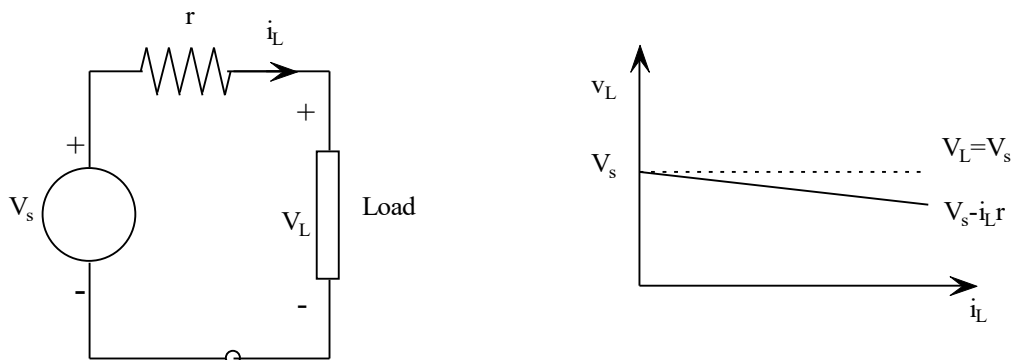


Fig 1.5 Practical Voltage Source and its characteristic

Practically every voltage source has small internal resistance shown in series with the voltage source and represented as ' r '. The voltage across the load $V_L = V_s - i_L r$.

The approximate internal resistance of a regulated power supply (RPS) is very low, typically less than 0.1 ohms ($m\Omega$).

2a. Ideal Current Sources: Ideal current source is defined as the energy source which gives constant current at its terminals irrespective of the voltage appearing across its terminals as shown in Fig 1.6.

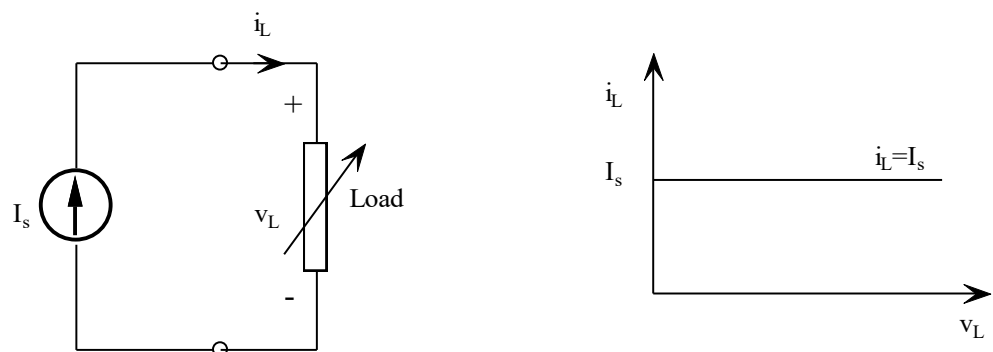


Fig 1.6 Ideal Current Source and its characteristic

2b. Practical Current source: In a practical current source, the current through the load keeps

falling as the voltage across the load increases as shown in Fig 1.7.

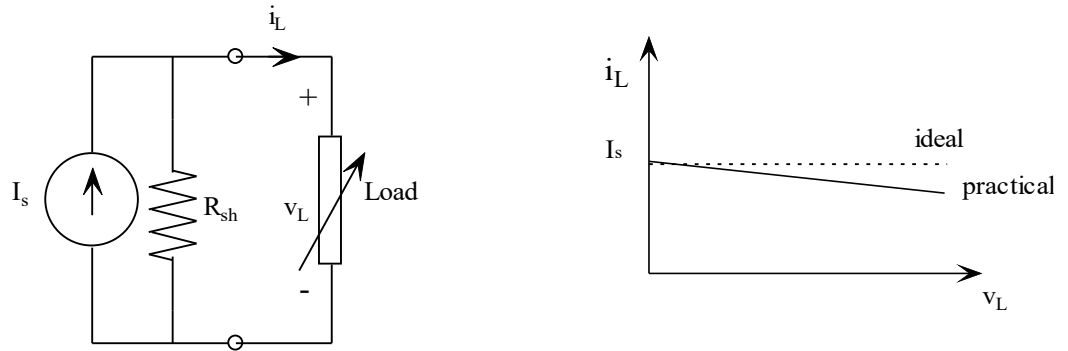


Fig 1.7 Practical Voltage Source and its characteristic

Practically every current source has high internal resistance in Mega Ohms shown in parallel with the current source and represented as ' R_{sh} '.

Resistances in Series: Two or more resistors are said to be connected in series when the same amount of current flows through all the resistors. In such circuits, the voltage across each resistor is different. The sum of the voltages across each resistor is the applied voltage. Consider the Fig 1.8 shown below

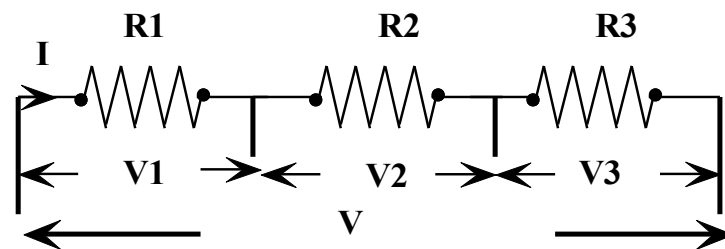


Fig 1.8 Resistances in Series

$$V=V_1+V_2+V_3$$

If $R_1=100\ \Omega$, $R_2=50\ \Omega$, $R_3=150\ \Omega$ then total resistance of the circuit=

$$R=R_1+R_2+R_3=100+50+150= 200\Omega$$

In a series connection, if any resistor is broken or a fault occurs, then the entire circuit is turned off.

Resistances in Parallel: Two or more resistors are said to be connected in parallel when the voltage is the same across all the resistors. In such circuits, the current through each resistor is different and the sum of the currents flowing through each resistor is the total current supplied by the source. Consider the Fig 1.9 shown below

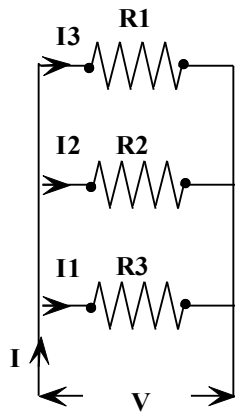


Fig 1.9 Resistances in Parallel

$$I = I_1 + I_2 + I_3$$

If $R_1 = 100 \, \Omega$, $R_2 = 50 \, \Omega$, $R_3 = 150 \, \Omega$ then total resistance of the circuit =

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{100} + \frac{1}{50} + \frac{1}{150} = 0.0367 \, \text{S}$$

$$R = 27.25 \, \Omega$$

Reciprocal of resistance is conductance and its unit is Siemens (mho, S).

A resistor or any other component can be connected or disconnected easily without affecting other elements in a parallel circuit.

Exercise 1: Find the overall resistance of the circuit shown in Fig 1.10

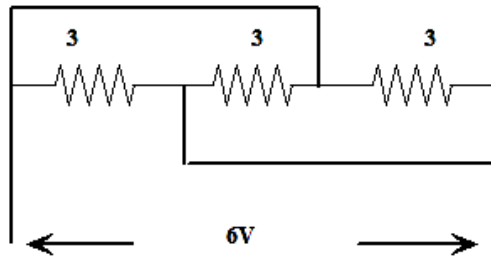
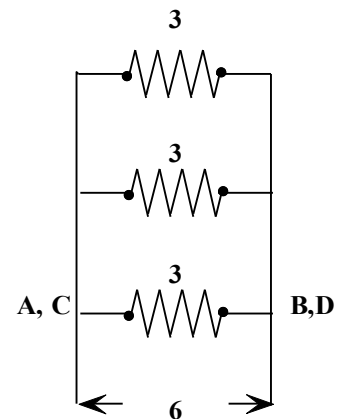
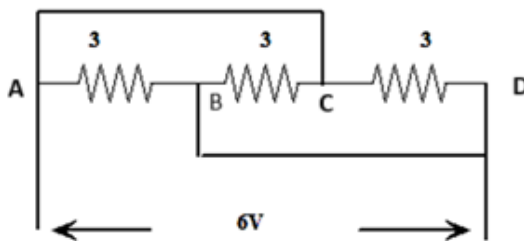


Fig 1.10 Exercise 1 (all resistances are in Ω)



$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1 \, \text{S}$$

$$R = 1 \, \Omega$$

Exercise 2: Find the overall resistance seen by the terminals of a network shown in Fig 1.11

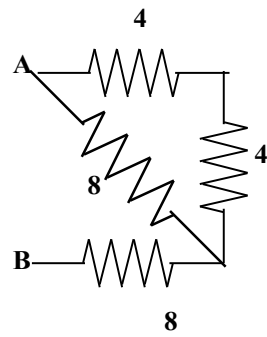
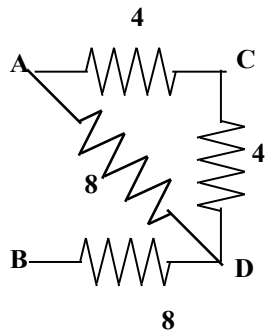
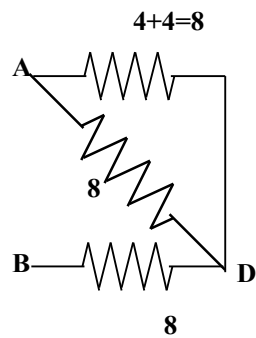


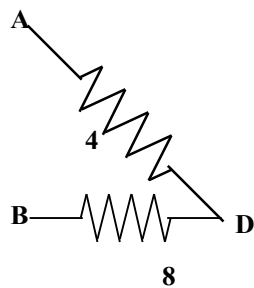
Fig 1.11 Exercise 2 (all resistances are in Ω)



4 Ω resistance between AC and CD are in series



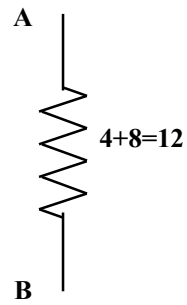
8 Ω resistances between AD are in parallel



$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{8} + \frac{1}{8} = 0.25 \text{ U}$$

$$R = 4 \Omega$$

4 Ω resistance between AD and 8 Ω between DB are in series



$$R_{AB} = 12 \, \Omega$$

Exercise 3: Find the overall resistance seen by the terminals AB of a network shown in Fig 1.12

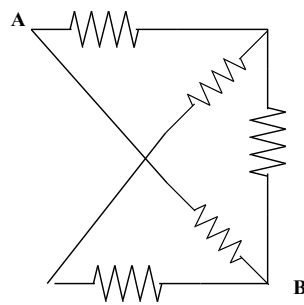
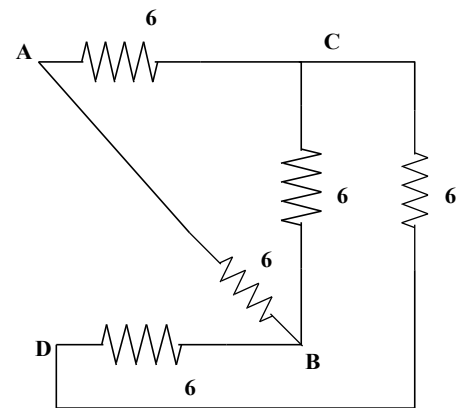
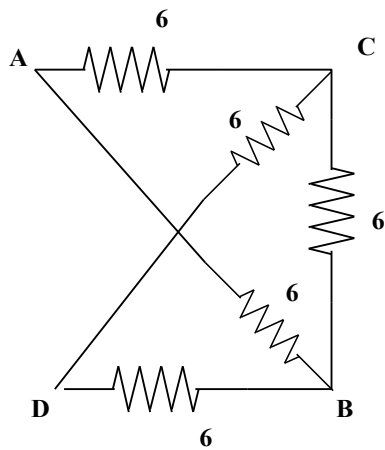
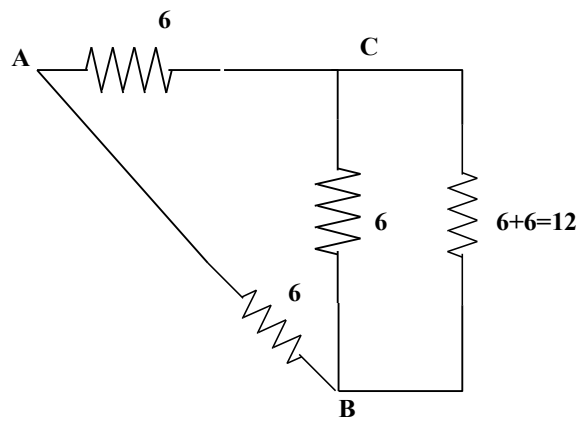


Fig 1.12 Exercise 3 (all resistances are of $6 \, \Omega$)

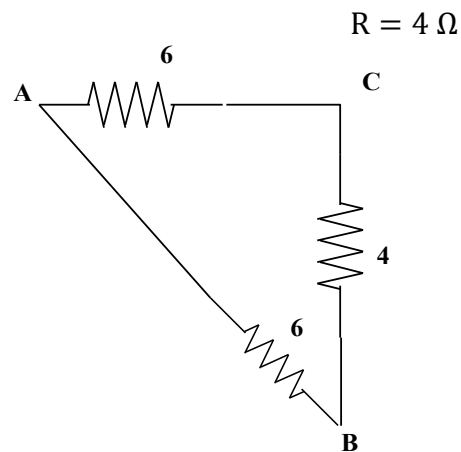


$6 \, \Omega$ resistance between BD and DC are in series

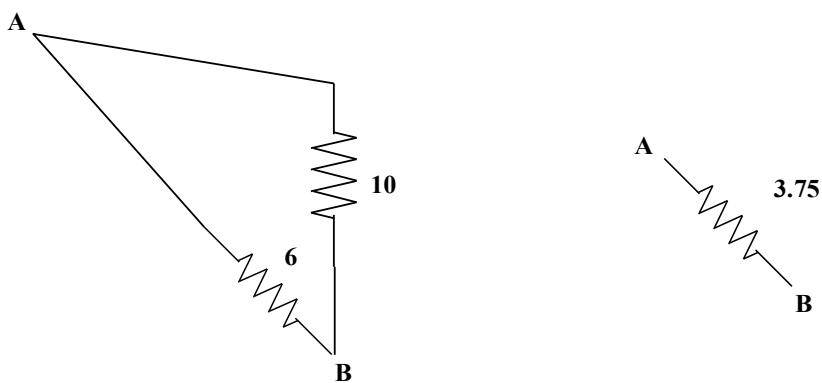


$6\ \Omega$ and $12\ \Omega$ are in parallel

$$\frac{1}{R} = \frac{1}{6} + \frac{1}{12} = 0.25\ \text{U}$$



$6\ \Omega$ between AC and AB and $4\ \Omega$ between CB are in series



$$R_{AB} = 3.75\ \Omega$$

Exercise 3: Find the overall resistance seen by the terminals of a network shown in Fig 1.13

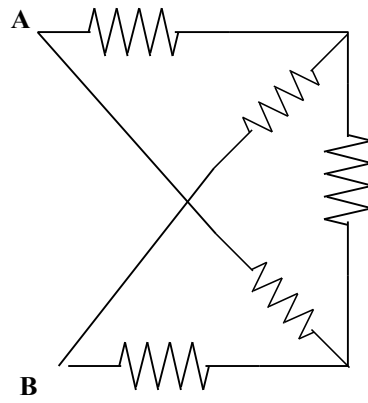
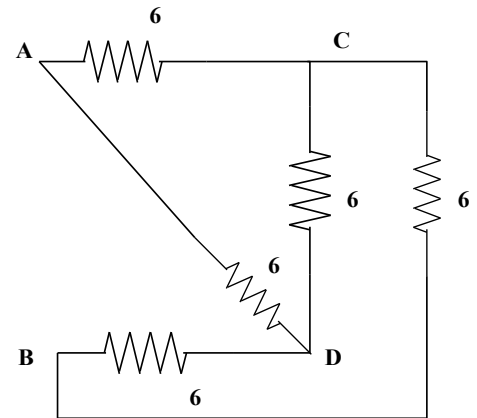
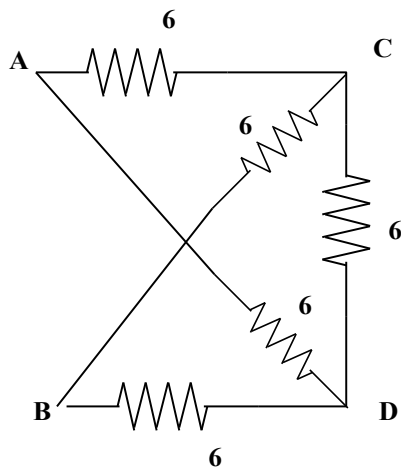


Fig 1.13 Exercise 3 (all resistances are of $6\ \Omega$)



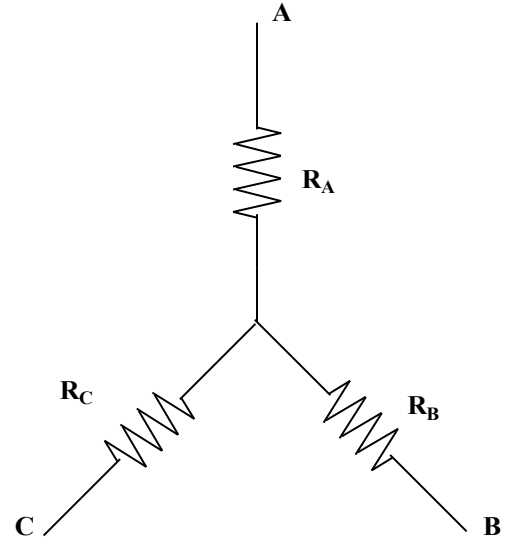
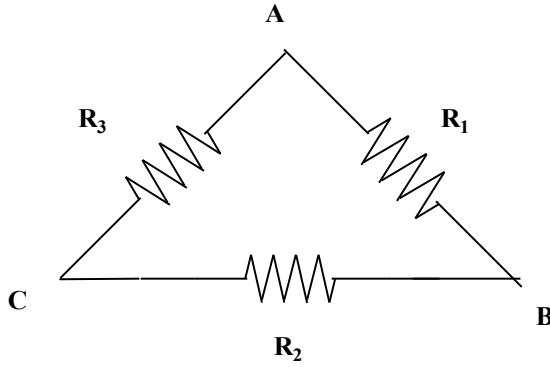
Further solution of the network is possible only by star-delta transformation

Network reduction using star delta transformation:

Certain network configurations such as above cannot be resolved by series-parallel combination. Such configurations are handled by Y- Δ transformation

Delta - Star Transformation

Consider a delta system that's three corner points are A, B and C as shown in the figure. Electrical resistance of the branches between points A and B, B and C and C and A are R_1 , R_2 and R_3 respectively



The resistance between the points A and B will be, $R_{AB} = R_1 \parallel (R_2 + R_3) = \frac{R_1(R_2+R_3)}{R_1+R_2+R_3}$

Analogues, star system connected to these points A, B, and C is as shown in the figure. Three arms R_A , R_B and R_C of the star system are connected with A, B and C respectively. Now if we measure the resistance value between points A and B, we will get, $R_{AB} = R_A + R_B$

Since the two systems are identical, resistance measured between terminals A and B in both systems must be equal.

$$R_{AB} = \frac{R_1(R_2 + R_3)}{R_1 + R_2 + R_3} = R_A + R_B \text{ ----- 1.1}$$

Similarly, resistance between points B and C being equal in the two systems,

$$R_{BC} = \frac{R_2(R_3 + R_1)}{R_1 + R_2 + R_3} = R_B + R_C \text{ ----- 1.2}$$

And resistance between points C and A being equal in the two systems,

$$R_{CA} = \frac{R_3(R_1 + R_2)}{R_1 + R_2 + R_3} = R_C + R_A \text{ ----- 1.3}$$

Adding equations 1.1, 1.2 and 1.3, we get,

$$2(R_A + R_B + R_C) = \frac{2(R_1R_2 + R_2R_3 + R_3R_1)}{R_1 + R_2 + R_3} \text{ ----- 1.4}$$

$$(R_A + R_B + R_C) = \frac{(R_1R_2 + R_2R_3 + R_3R_1)}{R_1 + R_2 + R_3} \text{ ----- 1.5}$$

Subtracting equation (1.2), from equation (1.5) we get,

$$R_A = \frac{R_3R_1}{R_1 + R_2 + R_3} \text{ ----- 1.6}$$

Similarly, subtracting equation (1.3), from equation (1.5) we get,

$$R_B = \frac{R_2R_1}{R_1 + R_2 + R_3} \text{ ----- 1.7}$$

Similarly, subtracting equation (1.1), from equation (1.5) we get,

$$R_C = \frac{R_2 R_3}{R_1 + R_2 + R_3} \text{-----1.8}$$

Star - Delta Transformation

For **star - delta transformation** we just multiply equations (1.6), (1.7) and (1.7), (1.8) and (1.8), (1.6) and then by adding we get,

$$\begin{aligned} R_A R_B + R_B R_C + R_C R_A &= \frac{R_1 R_2^2 R_3 + R_1 R_2 R_3^2 + R_1^2 R_2 R_3}{(R_1 + R_2 + R_3)^2} = \frac{R_1 R_2 R_3 (R_1 + R_2 + R_3)}{(R_1 + R_2 + R_3)^2} \\ &= \frac{R_1 R_2 R_3}{R_1 + R_2 + R_3} \text{----- 1.9} \end{aligned}$$

Now dividing equation (1.9) by equations (1.8), (1.6) and (1.7) separately we get,

$$\begin{aligned} R_1 &= \frac{R_A R_B + R_B R_C + R_C R_A}{R_B} = R_A + R_C + \frac{R_A R_C}{R_B} \\ R_2 &= \frac{R_A R_B + R_B R_C + R_C R_A}{R_C} = R_A + R_B + \frac{R_A R_B}{R_C} \\ R_3 &= \frac{R_A R_B + R_B R_C + R_C R_A}{R_A} = R_B + R_C + \frac{R_B R_C}{R_A} \end{aligned}$$

Exercise 3: Find the overall resistance seen by the terminals of a network shown in Fig 1.13

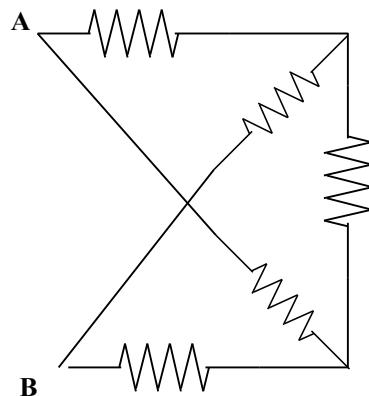
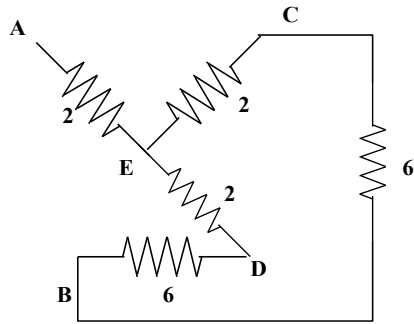
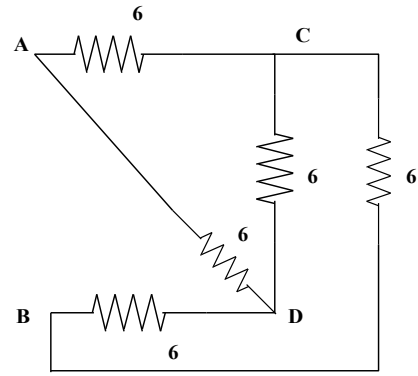
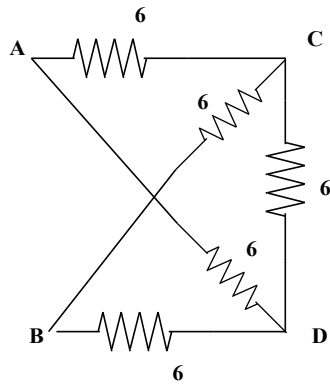
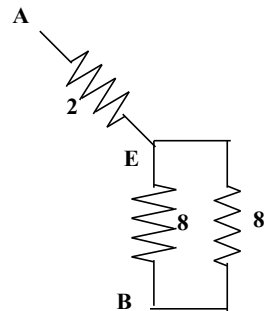


Fig 1.13 Exercise 3 (all resistances are of 6 Ω)

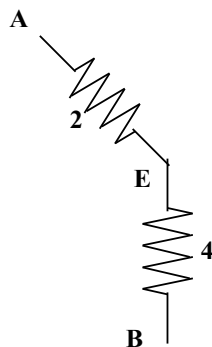


$2\ \Omega$ between EC and $6\ \Omega$ between CB are in series

$2\ \Omega$ between ED and $6\ \Omega$ between DB are in series



$8\ \Omega$ resistances between EB are in parallel



$2\ \Omega$ resistance between AE and $4\ \Omega$ resistance between EB are in series. Thus $R_{AB} = 6\ \Omega$

Exercise 4: Find the overall resistance of the network shown in Fig 1.14 seen between the terminals AB

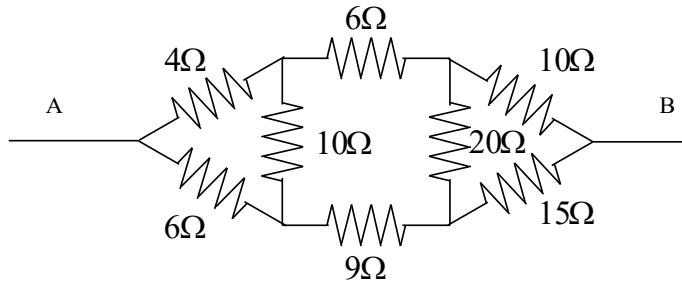
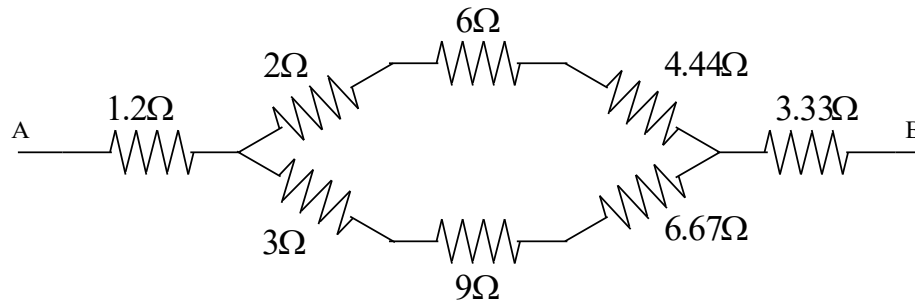
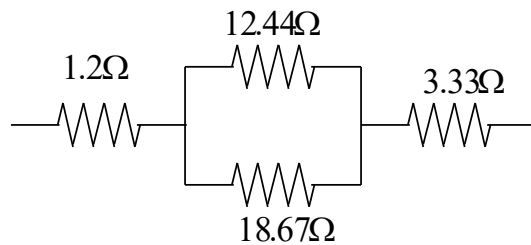


Fig 1.14 Exercise 4

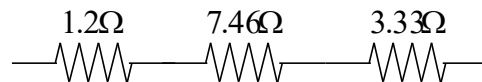
Step 1:



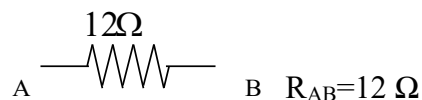
Step 2:



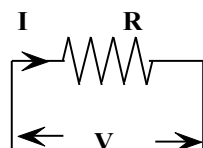
Step 3:



Step 4:



Ohms Law: Ohm's law states that the voltage across a conductor is directly proportional to the current flowing through it, provided all physical conditions and temperatures remain constant.



$$I = \frac{V}{R} \quad \text{where } R = \rho \frac{l}{a} \quad l = \text{length of material}$$

a = cross sectional area

ρ = Resistivity of the material

Exercise 5: Determine v_x by voltage division method for the circuit of Fig 1.15

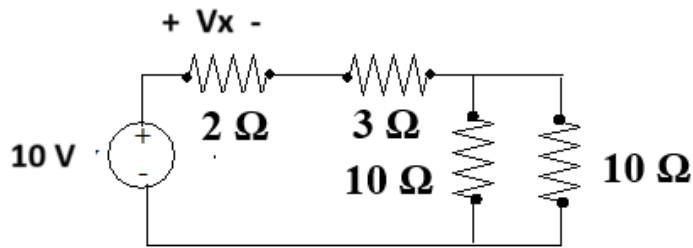
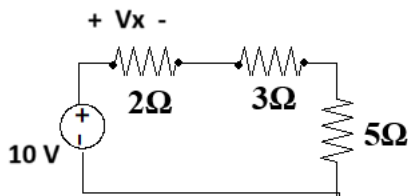


Fig 1.15 Exercise 5

Step 1: 10 Ω resistances are in parallel



Step 2: All three resistances are in series. Hence current I , $I = \frac{10}{2+3+5}$

The voltage v_x across 2Ω resistor is given by $v_x = \frac{10}{2+3+5} \times 2 = 2V$

Exercise 6: Determine I_1 and I_2 in the circuit of Fig 1.16 by current division method

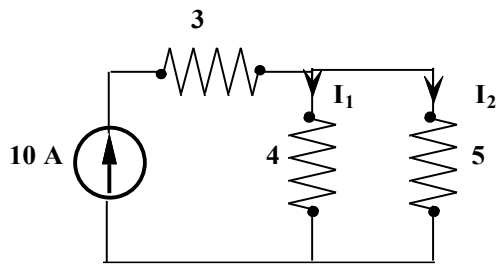


Fig 1.16 Exercise 6 (All resistances are in Ω)

Step 1: Current through 3 Ω resistance is 10 A

Step 2: Current division principle

$$I_1 = \frac{5}{4+5} \times 10 = 5.56 \text{ A}$$

$$I_2 = \frac{4}{4+5} \times 10 = 4.44$$

Source Transformation:

Two sources are said to be identical when their terminal voltage and load currents are same.

Consider a voltage source with internal resistance 'r' as shown in Fig 1.17

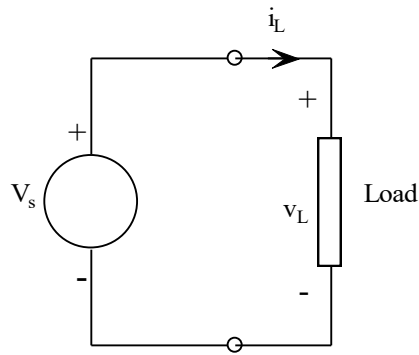


Fig 1.17 Voltage Source supplying a load

A voltage source with internal resistance 'r' can be converted into a current source, $I_s = \frac{V_s}{r}$ with internal resistance connected in parallel with it as shown in Fig 1.18

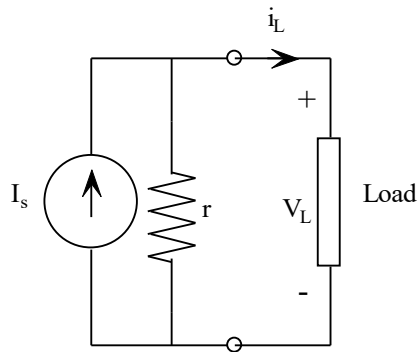


Fig 1.18 Current Source supplying a load

Likewise a current source I_s can be converted into a voltage source, $V_s = I_s R_{sh}$ with internal resistance 'R_{sh}' in series with it as shown in Fig 1.19

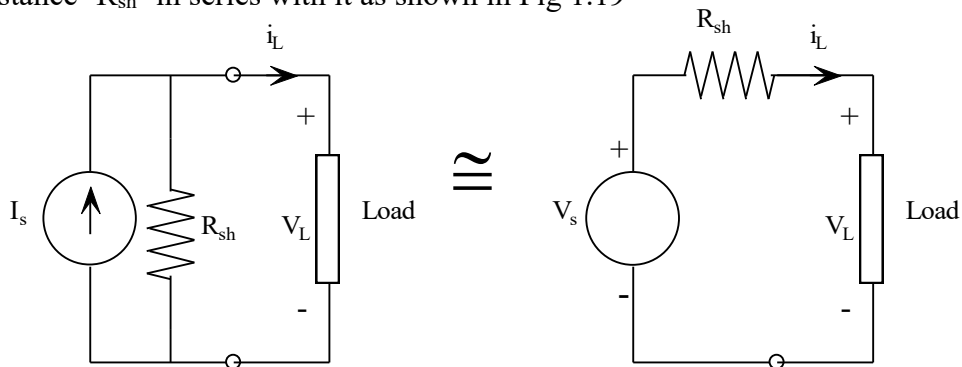


Fig 1.19 Current source to Voltage source Conversion

Exercise 7: Find the equivalent voltage source across AB in network shown in Fig 1.20 by using source transformation.

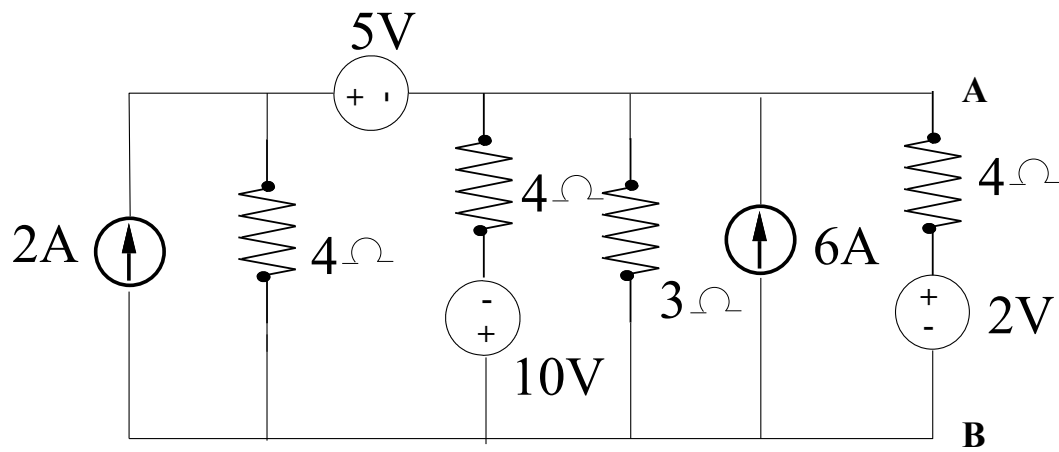
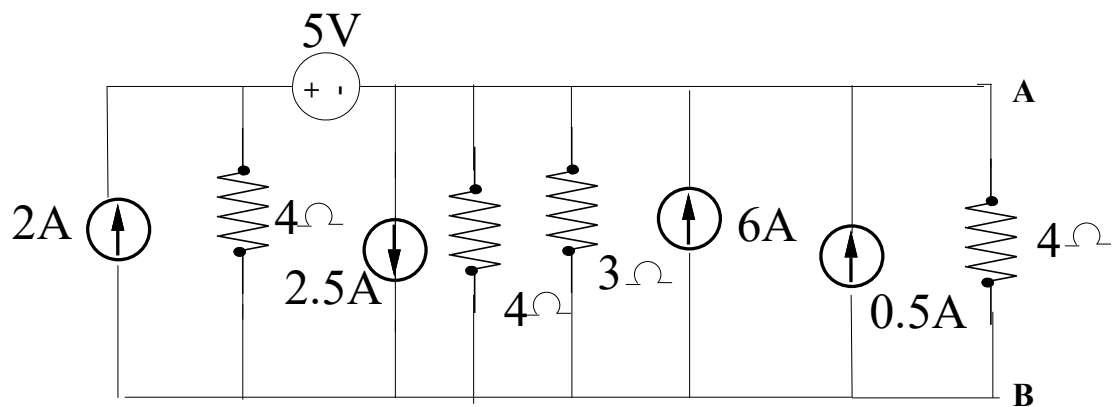


Fig 1.20 Exercise 7

Step 1: Convert all possible voltage sources into current sources

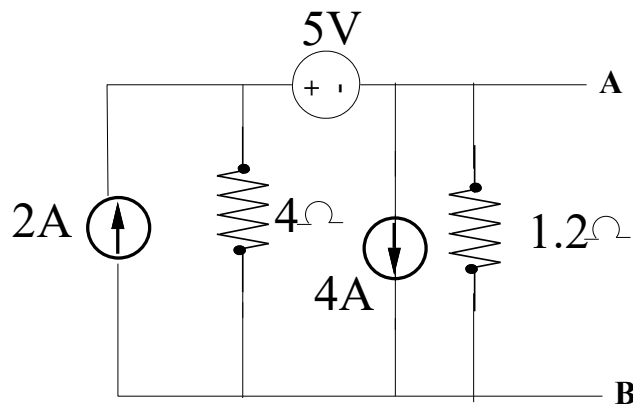


Step 2: 2.5A, 6A and 0.5 A sources are between the same nodes AB. So they can be replaced by a single current source $6+0.5-2.5=4A$.

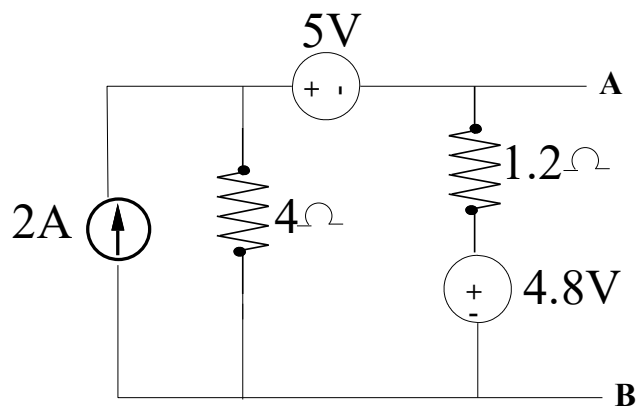
4Ω, 3Ω and 4Ω resistances are in parallel and can be replaced by a single resistance $\frac{1}{4} + \frac{1}{3} + \frac{1}{4} =$

$$\frac{1}{R_{eq}}$$

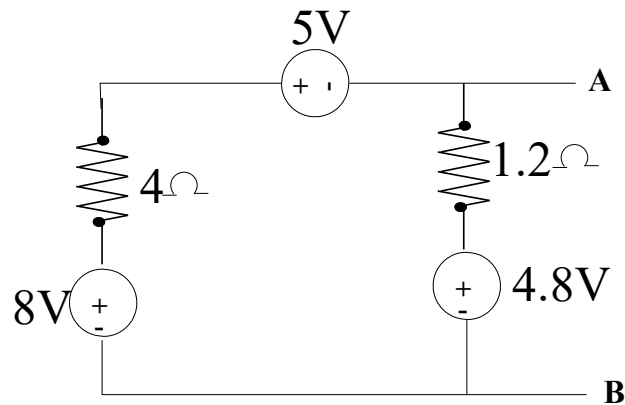
$$R_{eq}=1.2\Omega$$



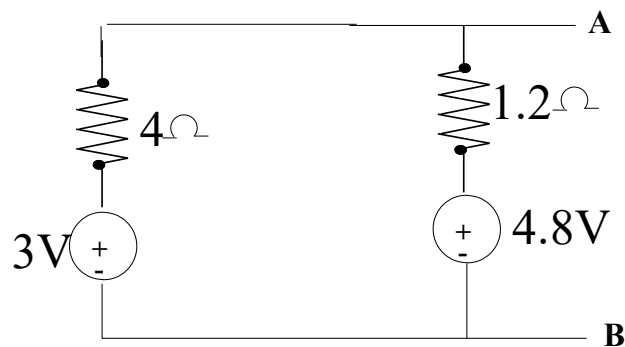
Step 3: Convert 4A current source in parallel with 1.2 Ω into a voltage source



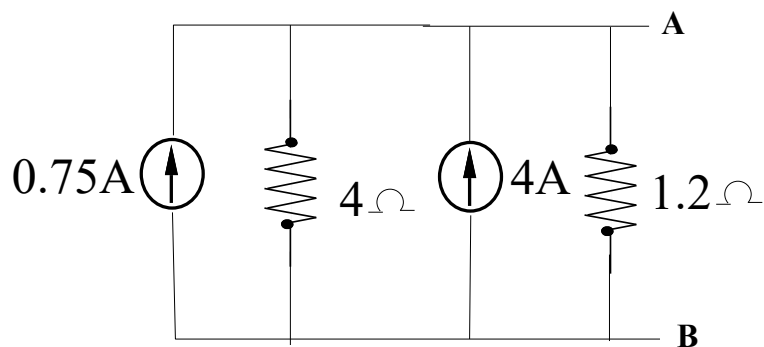
Step 4: Convert 2A current source in parallel with 4Ω into a voltage source



Step 5: 8V and 5V voltage sources are in series. So can be replaced by single voltage source $8 - 5 = 3V$



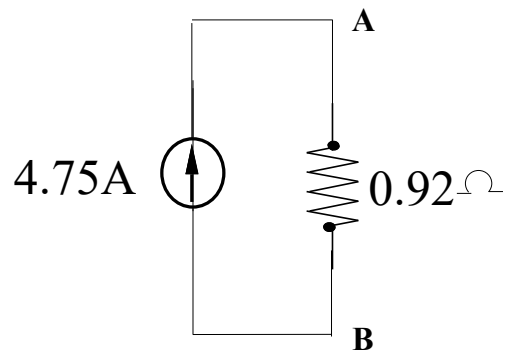
Step 6: Convert 3V and 4.8V voltage sources into equivalent current sources



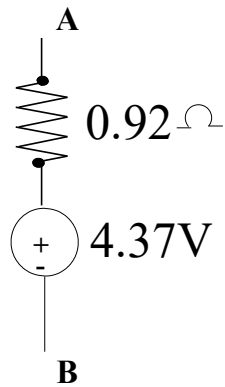
Step 7: 4 A and 0.75A current sources are between the nodes A and B. Hence they can be replaced by a single current source $4 + 0.75 = 4.75A$

4Ω and 1.2Ω resistances are in parallel and can be replaced by a single resistance $\frac{1}{4} + \frac{1}{1.2} = \frac{1}{R_{eq}}$

$$R_{eq}=0.923\Omega$$



Step 8: Convert 4.75A current source into a voltage source.



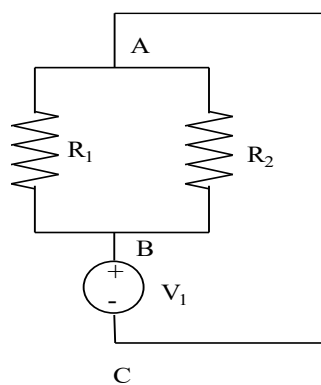
$$V_{AB}=4.37V$$

Source Shifting:

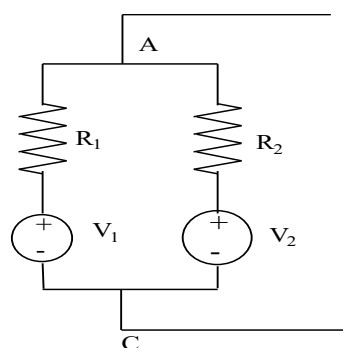
Whenever there is no resistance in series with the voltage source and no resistance in parallel with current source, then source transformation is not possible.

Voltage Source Shifting: E Shift method

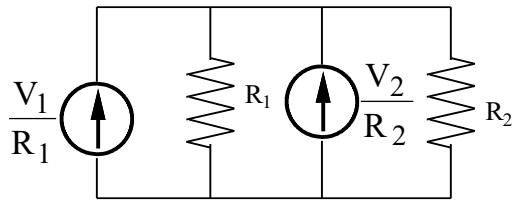
Consider the network shown below



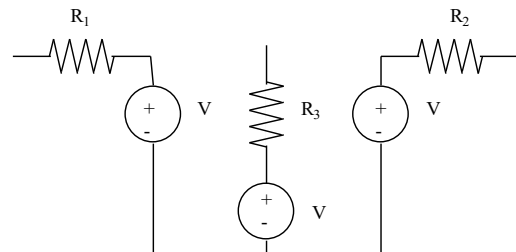
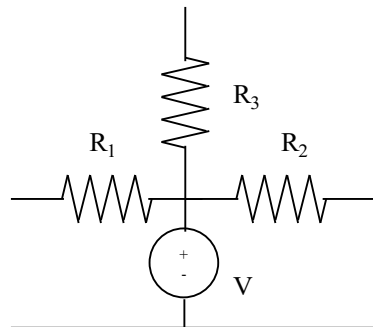
The above network can be rewritten as



This can be easily transformed into current source as shown



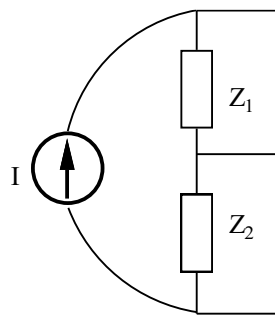
Note: while shifting a voltage source, the polarity of the voltage source should be taken care and must be marked correctly while shifting



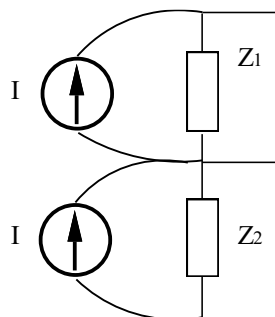
Current Source Shifting: I shifting

If in a network, there is no source impedance in parallel with a current source, then shifting of a current source is required.

Consider the network shown below

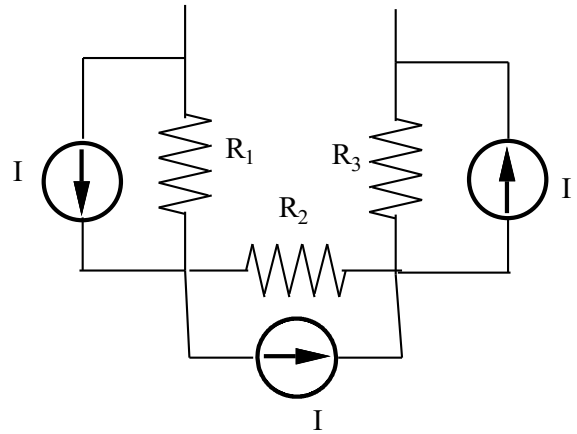
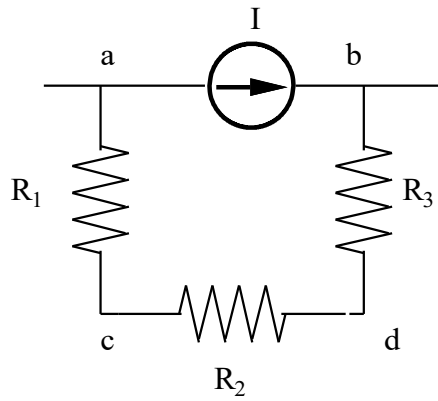


An additional identical current source is used in such a way that the currents at all the nodes remain unchanged.



Note: While shifting a current source, the current distribution at the nodes should not be affected.

For example,



Exercise 8: For the network shown in Fig 21, determine the voltage “V” using source shift or/and source transformation techniques only.

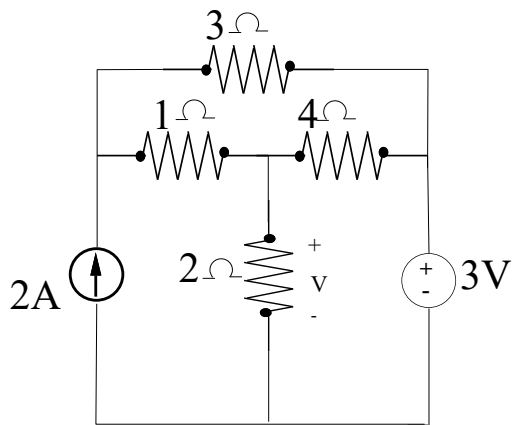
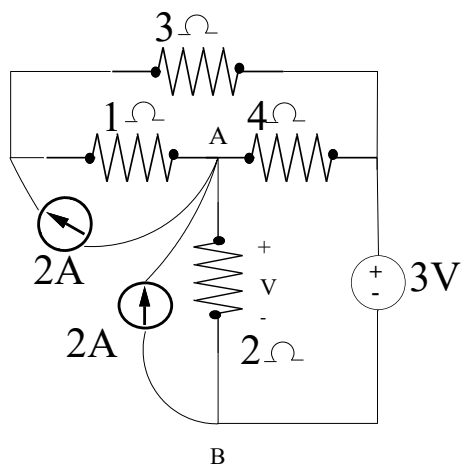
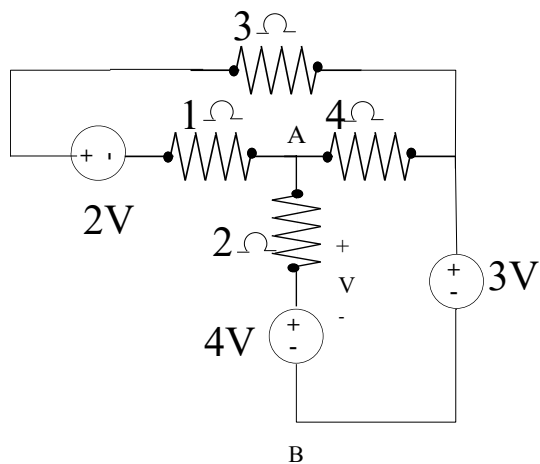


Fig 21 Exercise 8

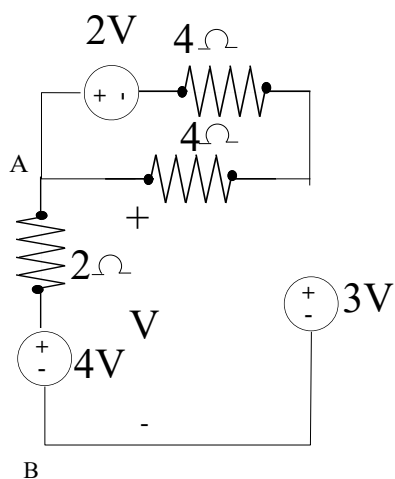
Step1: $2A$ current source can be split into 2 sources as shown below



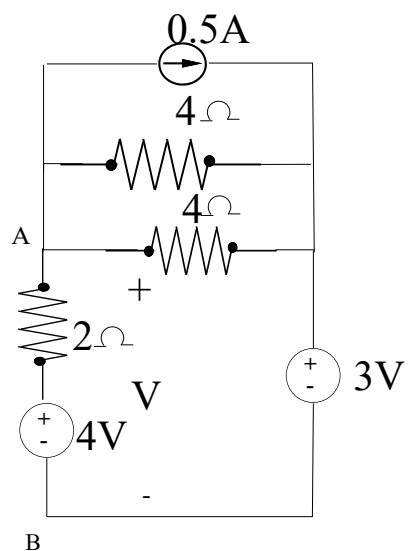
Step 2: Convert the current sources into equivalent voltage sources



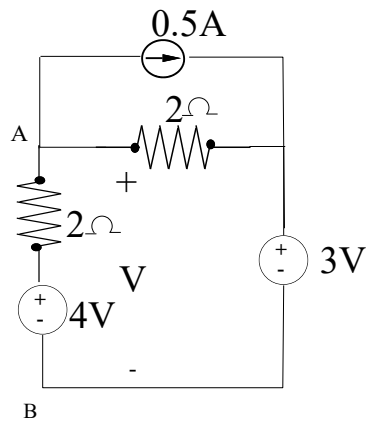
Step 3: 1Ω and 3Ω resistances are in series



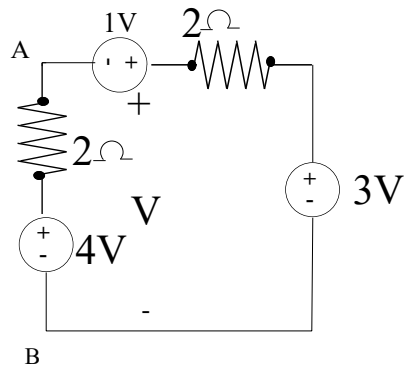
Step 4: Convert the 2V voltage source into current source



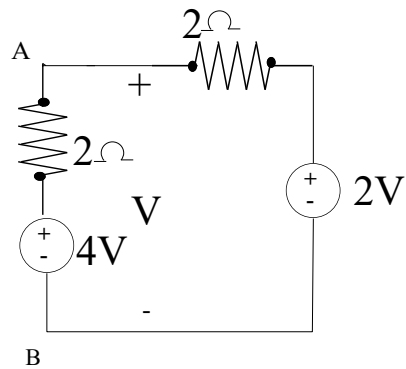
Step 5: 4Ω and 4Ω are in parallel



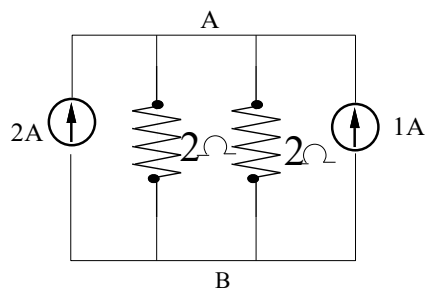
Step 6: Convert 0.5A current source into a voltage source



Step 7: 1V and 3V voltage sources are in series



Step 8: Convert the voltage sources into current sources

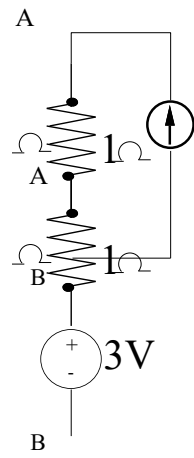


Step 9: 2A and 1A sources can be combined as $2+1=3A$

Two 2Ω resistances are in parallel, $R_{eq}=1\Omega$

Step

10:



3A Convert the current source into voltage source

$V_{AB}=3V$